



Semi-Supervised Online Continual Learning for 3D Object Detection in Mobile Robotics

Binhong Liu¹ · Dexin Yao¹ · Rui Yang² · Zhi Yan² · Tao Yang¹ 

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Abstract

Continual learning addresses the challenge of acquiring and retaining knowledge over time across multiple tasks and environments. Previous research primarily focuses on offline settings where models learn through increasing tasks from samples paired with ground truth annotations. In this work, we focus on an unsolved, challenging, yet practical scenario, specifically, the semi-supervised online continual learning in autonomous driving and mobile robotics. In our settings, models are tasked with learning new distributions from streaming unlabeled samples and performing 3D object detection as soon as the LiDAR point cloud arrives. Additionally, we conducted experiments on both the KITTI dataset, our newly built IUSL dataset and Canadian Adverse Driving Conditions (CADC) Dataset. The results indicate that our method achieves a balance between rapid adaptation and knowledge retention, showcasing its effectiveness in the dynamic and complex environment of autonomous driving and mobile robotics. The developed ROS packages and IUSL dataset will be publicly available at: <https://github.com/npu-ius-lab/OCL3D>.

Keywords Semi-supervised · Streaming learning · Online continual learning · 3D object detection

1 Introduction

In the past decade, autonomous driving and mobile robotics have experienced rapid development in both academia and industry [1]. Sensors provide observational data about the surrounding environment, which is then processed by perception algorithms, enabling the vehicle to understand and

respond to its surroundings more accurately. A perception system for autonomous vehicles consists of a combination of passive and active sensors, primarily cameras and LiDARs [2]. Cameras provide dense visual cues of color and texture information but lack precise ranging capabilities and are sensitive to changes in lighting conditions. LiDARs, on the other hand, provide accurate distance measurements even in varying lighting conditions but suffer performance degeneration in adverse weather.

Many state-of-the-art methods based on Deep Neural Networks (DNNs) for processing LiDAR point clouds have achieved considerable success in the realm of 3D object detection through offline training. However, these methods are incapable of adapting to changing environments. When autonomous vehicles operate in challenging weather conditions, e.g., dense fog, extreme snow, the performance of these DNN models trained under clear weather conditions may degrade significantly due to the shift in data domain [3]. A possible solution to this is to repeat the training process each time a domain shift occurs [4]. However, it's crucial to emphasize that retraining can be time-consuming and computationally expensive [5], and it also necessitates substantial amounts of expensive manual annotation data. Stream learning focuses on efficiently adapting to the current

✉ Tao Yang
yangtao@nwpu.edu.cn

Binhong Liu
binhongliu@mail.nwpu.edu.cn

Dexin Yao
736100452ydx@mail.nwpu.edu.cn

Rui Yang
rui.yang@utbm.fr

Zhi Yan
zhi.yan@utbm.fr

¹ Unmanned System Research Institute, National Key Laboratory of Unmanned Aerial Vehicle Technology, Integrated Research and Development Platform of Unmanned Aerial Vehicle Technology, Northwestern Polytechnical University, Xi'an 710072, China

² CIAD UMR7533, Univ. Bourgogne Franche-Comté, UTBM, 90010 Belfort, France

data domain without considering past data domains [6]. However, this emphasis on rapid adaptation can lead to catastrophic forgetting [7], where the model forgets previously learned knowledge when exposed to new information or domain shifts. In the context of overcoming these challenges, online continual learning emerges as a promising approach. It involves the model learning from a continuous stream of data without predefined task boundaries. This allows for the simultaneous evaluation of both information retention and online learning efficacy [8].

In consideration of these aspects, there is a clear need for an entirely new class of algorithms capable of achieving rapid adaptation in new domains and knowledge retention in old domains, with minimal or no dependence on manually annotated data in the realm of LiDAR-based 3D object detection in autonomous driving. In this work, we propose a method for 3D object detection that enables rapid adaptation to new domains, retains knowledge from old domains, and achieves this without the use of manually annotated data by leveraging a pre-trained 2D object detector.

This article is an extension of our previous work [9] and includes the following new contributions.

- We propose a semi-supervised online continual learning pipeline for 3D object detection in autonomous driving to gradually adapt to the changing environments without saving samples.
- We provide comprehensive experimental analyses, especially: extending the evaluation with both hand-crafted and deep learning features; comparing our proposed framework with the end-to-end method on the KITTI dataset; and conducting cross-domain experiments on our custom dataset, named IUSL dataset and CADC dataset [10].
- The end-to-end detectors, feature extractors, and online learning modules could be the plug-in modules integrated into our framework. The source code and IUSL dataset will be released to benefit the autonomous driving and robotics communities.

The rest of the paper is organized as follows: Section 2 introduces the related work, particularly on 3D object detection under different domains and online learning. Section 3 presents the details of our proposed framework. Section 4 shows the experimental settings and provides a comprehensive experimental evaluation of the proposed method. Finally, conclusions and future work are given in Section 5.

2 Related Work

2.1 3D Object Detection Crossing Domains

LiDAR-based 3D object detection methods [11–15] have made significant strides in the realm of autonomous driv-

ing due to the great success of DNNs. However, it's hard for models trained on the source domain to achieve satisfactory performance on the target domain due to the domain gap caused by changes in sensor types, e.g., 64-layer LiDARs and 32-layer LiDARs, weather conditions, e.g., clear day and foggy day, and environments, e.g., urban scenarios and country scenarios. To address this problem, domain adaptation techniques have been well-developed for the 3D cross-domain detectors [16–21] that handle the domain shifts in autonomous driving. ST3D [18] proposed a self-training method to adapt a pre-trained detector from the source domain to the target domain. Wei et al. [17] designed a LiDAR distillation framework to bridge the domain gap induced by different LiDAR layers for 3D object detection. Bi3D [16] developed a domains-aware source sampling strategy that helps the model select target-domain-like samples from the source domain. Using this strategy, the model can actively choose the most informative samples from the source domain to effectively adapt to the target domain. While these methods effectively mitigate performance degradation caused by domain shifts, it is essential to note that these approaches primarily focus on changes in sensor types and environments but seldom consider changes in LiDAR performance degeneration. Additionally, these approaches still require offline training, leading to substantial consumption of computational resources and be not able to deploy in-situ.

2.2 Online Streaming Continual Learning

Online learning (OL) represents a family of machine learning methods, where a learner attempts to tackle some predictive or decision-making tasks by learning from a sequence of data instances one by one at each time [22]. Unlike traditional machine learning, online machine learning means that the model can update across different domains over time spontaneously and automatically after deployment in the real world [23]. Streaming Learning (SL) focuses on adapting to data distribution quickly without the additional learning objective of preserving old knowledge explicitly [6]. The settings of SL are: to predict at any given moment, dynamically adapt to underlying data distribution changes (concept drifts), and be computationally efficient when learning and predicting [24]. Online Continual Learning (OCL) is a subproblem in OL that models learn over time from a continuous stream of data without having to revisit previously encountered training samples and can preserve old knowledge [25]. Online Streaming Continual Learning (OSCL) is identified as the intersection of SL and OCL [6]. In OSCL, the model faces the dual challenge of adapting to dynamic changes in data distribution (i.e., SL) while also ensuring the preservation and accumulation of knowledge over time (i.e., OCL). Like the traditional machine learning methods, the model also requires supervisory signals when the model

updates in OSCL literature. However, manual annotation of data becomes challenging after deployment in the real world. In recent years, research on OL for mobile robots to update models in situ and on the fly has become more and more extensive [26–30]. Teichman et al. [29] designed a semi-supervised method for track classification in dense 3D range data. By initiating the learning process with a small set of hand-labeled seed trajectories, the proposed method achieves results comparable to those of a fully-supervised method. This highlights the effectiveness of a method based on the motion trajectories of targets in generating reliable learning samples. Yan et al. [26–28] proposed an online self-supervised learning framework for human classification and an online transfer learning framework for human detection by using an online label generator module based on multi-object tracking. These frameworks allow the model to update spontaneously after deployment. Yang et al. [30] developed an efficient online transfer learning framework by employing an offline-trained 2D image detector for online labeling of 3D LiDAR point clouds, enabling the model to update spontaneously and automatically. This framework emphasizes rapid domain adaptation, especially when the model needs to adapt to different environments. Our research further focuses on both rapid domain adaptation to changes in LiDAR performance affected by weather conditions, i.e., from clear to foggy and from light snow to extreme snow, and knowledge retention.

The knowledge transferring from 2D detector to 3D detector is performed on the fly without saving samples. This distinctive focus categorizes our research within the domain of online continual learning for streaming data in a semi-supervised learning fashion.

Several methods for the evaluation of a model are explained in the Streaming Learning literature [6]. The most popular one is the *test-then-train* mode [6, 24] which means the model tests every incoming data and later be trained. However, in the *test-then-train* mode, the evaluation primarily focuses on assessing the model's domain adaptation performance, but it falls short in evaluating the model's knowledge retention performance. To assess the model's knowledge retention performance, we allow our model to learn from LiDAR point clouds with different data quality affected by weather conditions changing, i.e., from clear to foggy and from light snow to extreme snow, and then evaluate the model back to clear weather or light snow. This evaluation mode is identified as *test-after-train*.

3 Proposed Method

In this section, we explain the theory and implementation of our approach. Figure 1 provides a detailed module diagram of the framework. Our method comprises three main components: an online-trained 3D object detector, a pseudo

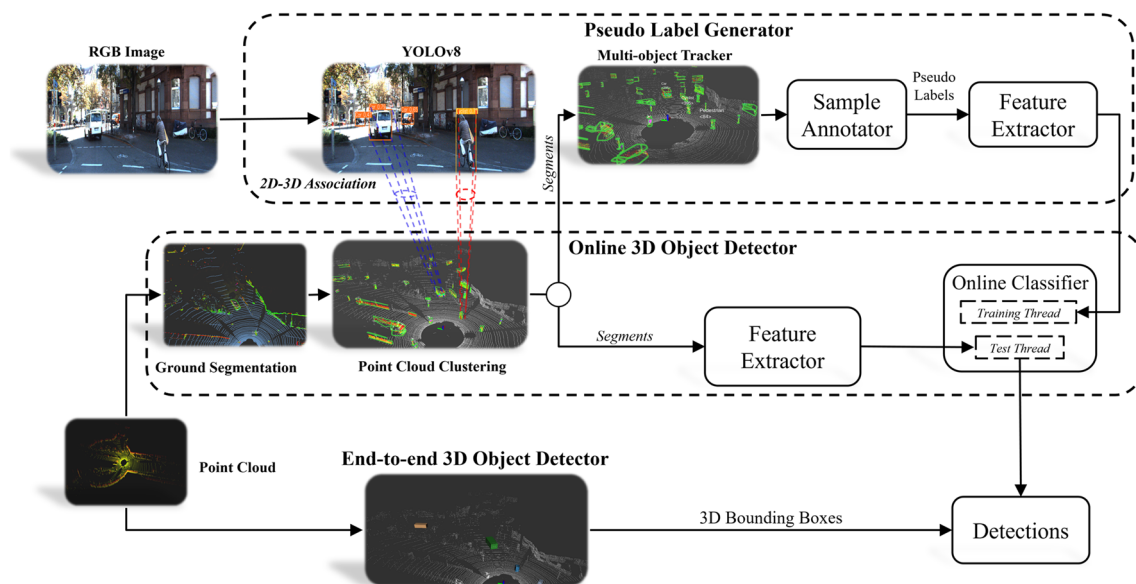


Fig. 1 The framework for semi-supervised online continual learning in 3D object detection for mobile robotics primarily includes three main components: an online-trained 3D object detector, a pseudo label generator, and an end-to-end 3D object detector. The online 3D object detector consists of four parts: ground segmentation, point cloud cluster-

ing, feature extractor, and online classifier. The pseudo label generator comprises 2D-3D association, a multi-object tracker, and a sample annotator. Additionally, an offline-trained end-to-end 3D object detector can be optionally integrated as a plug-and-play module within our framework

label generator, and an end-to-end 3D object detector. The online 3D object detector is specifically designed to effectively handle environmental changes, adapt seamlessly to different LiDAR performance, and mitigate catastrophic forgetting when learning across diverse domains.

The pseudo label generator collaborates with the online 3D object detector by generating accurate learning samples without the dependency on manual annotations. In addition, an offline-trained end-to-end 3D object detector can be a plug-in module in the post-processing fusion. Since the high-quality pseudo labels for 3D detection are generated from the knowledge of a pre-trained 2D object detector and object tracking sequences, YOLOv8 [31] is chosen for a balance between accuracy and processing speed. However, as YOLOv8 was trained on the COCO dataset [32], it does not include the classes of “pedestrian” and “cyclist” as defined in the KITTI dataset. Therefore, in our use, the class of “person” output by YOLOv8 is mapped to “pedestrian”, while a “person” with a “bicycle” below is mapped to “cyclist” class.

3.1 Online 3D Object Detector

The online 3D object detector consists of four components: ground segmentation, point cloud clustering, feature extractor, and online classifier.

Patchwork++ [33] is used because of its fast and accurate ground segmentation performance, as shown in Fig. 2. Then, we utilize the clustering method from Autoware [34] which implicitly projects points onto a 2D (x-y) plane and segments according to Euclidean distance with cluster output represented by an Oriented Bounding Boxes (OBB).

Both hand-crafted features and deep learning features are tested. For hand-crafted features, each cluster forms a 61-dimensional feature vector $f_j = (f_1, \dots, f_6)$ input to the online classifier [30]. Details are illustrated in Table 1.

We employ PointNet [35] as the deep learning feature extractor which has a 1024-dimensional feature vector of output logits. To align with the dimension of the hand-crafted

Table 1 Hand-crafted features

Feature	Description	Dimension
f_1	Number of points in the cluster	1
f_2	Minimum distance to the sensor	1
f_3	Covariance matrix of the cluster	6
f_4	Moment of inertia tensor	6
f_5	Slice feature for the cluster	20
f_6	Reflection intensity distribution	27

features, a fully connected layer with 61-dimensional output is connected to the layer of logits as the final output features. More recent point cloud networks, i.e., PointNet++ [36] and PointNN [37], are also tested. However, PointNet++ is less efficient especially when processing multiple objects, and PointNN needs a fixed number of input points which doesn't fit our scenarios. Figure 3 illustrates the PointNet feature extractor.

Cluster features are finally fed to a classifier trained online, where Aggregated Mondrian Forests (AMF) is applied. AMF is an online classifier that can learn new classes and distributions from streaming data and make classifications when a new sample is coming [38]. The classic Random Forest (RF), Support Vector Machine (SVM), and Neural Network (NN) exclusively support training from saved batch data, which does not meet the fashion of online learning from stream data. Figure 4 illustrates the process of training and classification performed by our online learning module.

3.2 Pseudo Label Generator

The key to enabling semi-supervised online learning in our framework is the reliable generation of pseudo labels. The Pseudo Label Generator consists of 2D-3D association, multi-object tracker, and sample annotator.

The 2D-3D association is to associate the 2D bounding boxes output by YOLOv8 and the OBBs output by point

Fig. 2 Ground segmentation results using Patchwork++

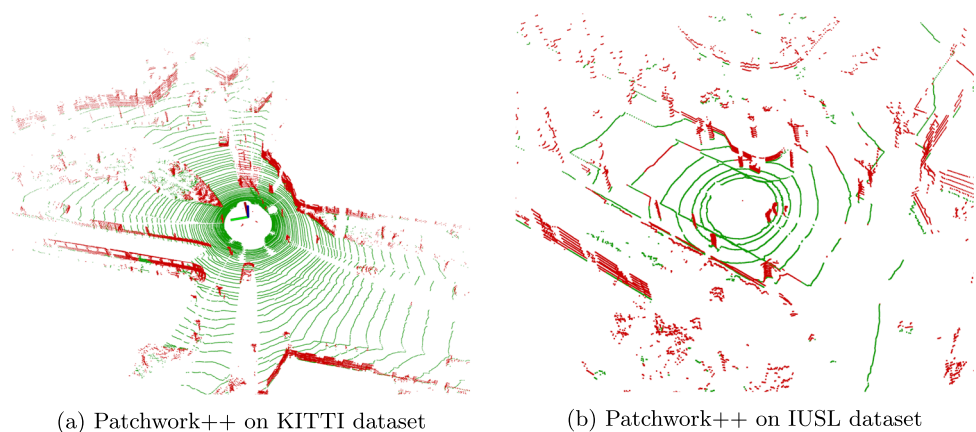
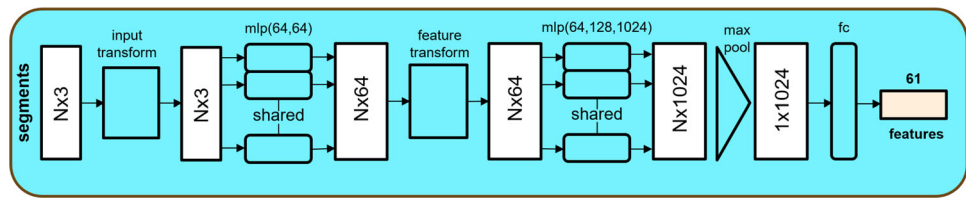


Fig. 3 Feature extraction module based on PointNet encoder and a fully connected layer



cloud clustering. The association is implemented by projecting the 3D OBBs to the 2D image plane using the Intersection over Union (IoU) between the projected 2D bounding boxes and 2D bounding boxes from YOLOv8 with a predefined threshold. Therefore, the output from the 2D detector is assigned to the OBBs from the point cloud in the front view.

The simple projecting association may encounter issues when an object is partially occluded or moves outside of the camera's front view. These issues lead to wrong pseudo label associations and inefficient learning, respectively. To overcome these challenges, the Autoware multi-object tracker [34], which is based on IMM (Interacting Multiple Model), UKF (Unscented Kalman Filter), and PDA (Probabilistic Data Association), is utilized as a part of the pseudo label generator. The tracker associates OBBs from different time steps, which is able to reduce ambiguity caused by incorrect 2D-3D Association module and generate pseudo labels by trajectories. However, despite the tracker being able to associate OBBs across different time steps, it still encounters some errors, mainly ID switches and incorrect data associations, as shown in Fig. 5. Figure 5a depicts the ideal scenario, where the tracker successfully associates OBBs across time steps. In contrast, Fig. 5b demonstrates a case where ID switches occur, prematurely ending the tracking of an object, leading to shorter trajectories and affecting the generation of pseudo labels. When trajectories are too short, we discard them. Incorrect data associations can also have a significant impact on pseudo label generation. In crowded environments, the tracker is prone to such errors, as shown in

Fig. 5c. In these cases, the we ensures that each trajectory contains objects of only one class, thereby reducing the impact of incorrect data associations. The robust spatio-temporal association holding scenarios of crowded environments and missing detections is illustrated in Fig. 6.

We utilize track probability to determine whether the samples on the trajectory are reliable enough to be used as pseudo labels. Specifically, the track probability is denoted as $P(Y_T|X_T, D)$. All the OBBs associated with the object in the trajectory are represented as $X_T = \{x_1, x_2, \dots, x_t\}$, while the corresponding pseudo labels are denoted as $Y_T = \{y_1, y_2, \dots, y_t\}$.

The classification confidence of OBB from the 2D detector D at time i with class y_i is denoted as $P(y_i|x_i, D)$. The confidence of pseudo labels for OBBs across the entire trajectory can be denoted as $P(Y_T|X_T, D)$ and is calculated as follows [27]:

$$P(Y_T|X_T, D) = \frac{\text{odds}_{X_T}}{1 + \text{odds}_{X_T}} \quad (1)$$

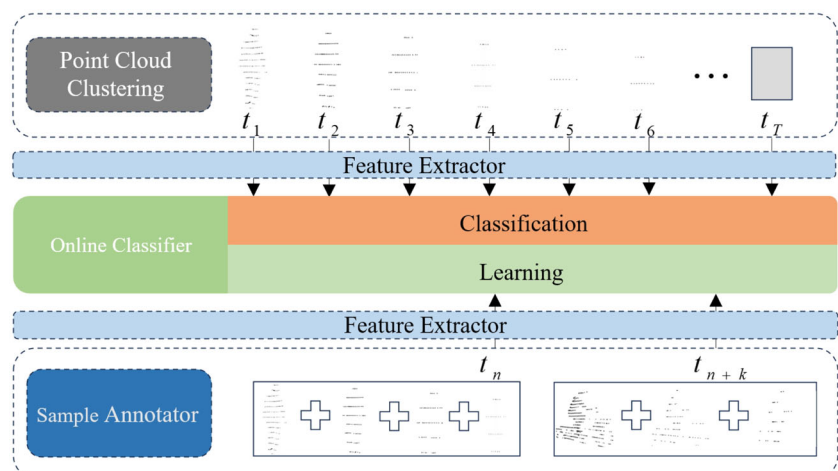
where

$$\text{odds}_{X_T} = \prod_{i=1}^t \text{odds}_{x_i} \quad (2)$$

and

$$\text{odds}_{x_i} = \frac{P(y_i|x_i, D)}{1 - P(y_i|x_i, D)} \quad (3)$$

Fig. 4 Schematic of the Online Classifier module working pipeline: LiDAR point cloud arrives at a certain rate, and the model performs classification. The online classifier updates based on examples paired with pseudo labels generated by the Sample Annotator module



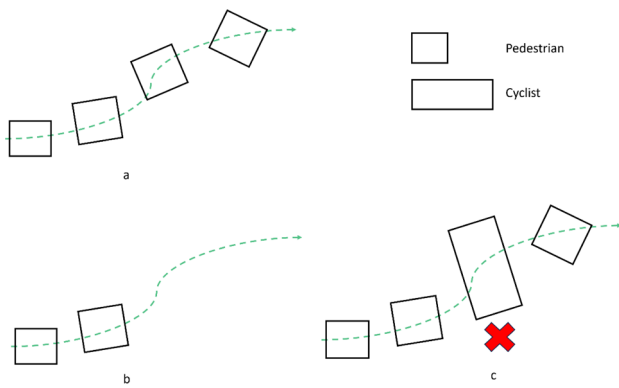


Fig. 5 Three different tracks. Figure (a) shows the ideal case, figure (b) shows the too short trajectory due to ID switch, and figure (c) showcase the incorrect data associations

By thresholding $P(Y_T|X_T, D)$, high-confidence samples are sent to the feature extractor module.

3.3 End-to-end 3D Object Detector

An offline-trained end-to-end 3D object detector could be an optional and plug-and-play module into our framework. PointPillars is chosen due to its high efficiency. The post-processing fusion is separated into two steps, i.e., 3D bounding box fusion and classification fusion, which is detailed in Algorithm 1.

Algorithm 1 Fusion strategy for online and end-to-end 3D object detectors.

Require: List of N OBBs from online 3D object detector, list of M OBBs from PointPillars, confidence threshold of PointPillars's OBBs $Conf_{thre}$, and IoU threshold IoU_{thre}

Ensure: Final detections OBBs with associated class information

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1: Final detections OBBs  $obb_{final}$ 
2: for each OBB  $obb_{online}$  of online 3D object detections do
3:   Initialize maximum IoU:  $maxIoU \leftarrow 0$ 
4:   Initialize matched OBB:  $matched\_obb \leftarrow \text{None}$ 
5:   for each OBB  $obb_{pillar}$  of PointPillars OBBs do
6:     if confidence of  $obb_{pillar} < Conf_{thre}$  then continue
7:   end if
8:    $IoU \leftarrow IoU(obb_{online}, obb_{pillar})$ 
9:   if  $IoU > maxIoU$  then
10:    Update  $maxIoU \leftarrow IoU$ 
11:    Update  $matched\_obb^1 \leftarrow obb_{pillar}$ 
12:   end if
13: end for
14: if  $maxIoU > IoU_{thre}$  then
15:   Add  $matched\_obb$  to  $obb_{final}$ 
16: else
17:   Add  $obb_{online}$  to  $obb_{final}$ 
18: end if
19: end for
20: Add all the unmatched  $obb_{pillar}$  to  $obb_{final}$ , if confidence of  $obb_{pillar} > Conf_{thre}$ 

```

¹ Note that the class information of $matched_obb$ determined by the confidence scores from both detectors.

An important motive for integrating this optional and plug-and-play module into our framework and designing the fusion strategy is we want to leverage the distinctive strengths of both detectors, enhancing the system's detection performance. On one hand, the offline-trained end-to-end 3D object detector produces accurately 3D bounding boxes; On the other hand, the online 3D object detector can complements detections in cases where the offline detector misses detections, as the clustering-based online 3D object detector seldom misses detections.

4 Experiments

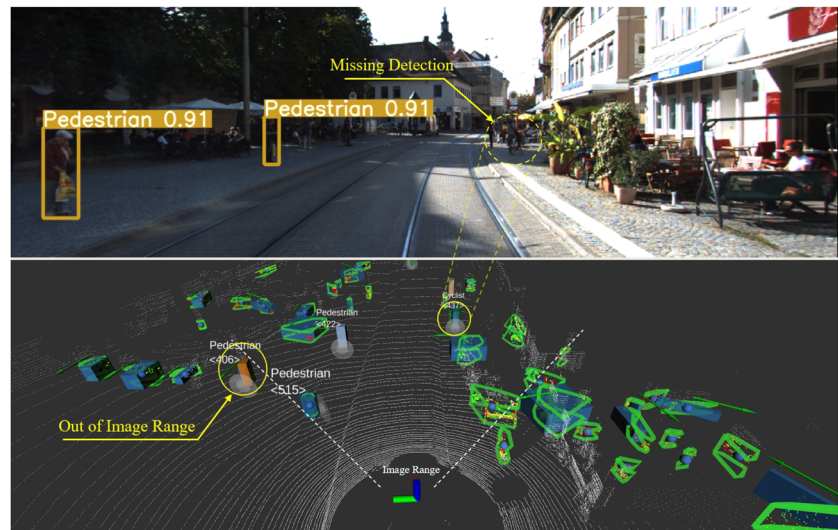
4.1 Experimental Settings

We evaluate our proposed method using the KITTI dataset, IUSL dataset and CADC dataset. The confusion matrix is used to evaluate point cloud cluster classification performance, and the Average Precision (AP) is used to assess the 3D object detection performance. The experiments are all based on the setting of semi-supervised online continual learning for streaming data, where learning is performed from the unlabeled raw data and the system is capable of running in real-time. All the experiments are conducted on Ubuntu 18.04 LTS (64-bit) with ROS Melodic, using an Intel i7-12700K CPU and NVIDIA GeForce RTX 3090 GPU.

KITTI dataset Since our method deals with continuous and unlabeled data, we use the KITTI 3D object tracking dataset for the evaluation. However, there are only a few sequences containing the objects of pedestrians and cyclists in the dataset. Therefore, we select two typical sequences for each class to perform *test-then-train* evaluation. For cars, we use the sequences 0006 and 0010. For pedestrians, we use the sequences 0016 and 0017. For cyclists, we use the sequences 0010 and 0012. Subsequently, we employ the data from the remaining sequences to train the PointNet feature extractor and PointPillars. $Conf_{thre}$ and IoU_{thre} in Algorithm 1 are set to 0.5 and 0.3, respectively. Track probability, denoted as $P(Y_T|X_T, D)$, is set at 0.7 to generate reliable samples. The IoUs of cars, pedestrians, and cyclists as set to 0.5, 0.25, and 0.25 respectively. The 3D detection results are filtered out for both the online learning detector and end-to-end detector by a confidence threshold $Filter_{thre}$ that is set to 0.5, because using a relatively low confidence threshold leads to a high number of false positives which isn't penalized by standard evaluation metrics [39].

IUSL Dataset We collected the IUSL dataset in the Youyi campus of Northwestern Polytechnical University using a multi-sensor mobile robot which was equipped with a

Fig. 6 OBBs associated with corresponding YOLOv8 2D detection boxes. The multi-object tracker assists in maintaining class information for OBBs when targets exit the camera's front view or YOLOv8 detections are temporarily lost



RoboSense 16-layer LiDAR, an RGB camera, an Oculii Eagle 4D radar, and a Livox Avia LiDAR, as illustrated in Fig. 7d. 4721 continuous soft time-synchronized frames of 16-layer LiDAR's point clouds and RGB images were extracted. The road participants, i.e., cars, pedestrians, and cyclists, within a 15m range in front of the LiDAR of 999 frames are manually labeled as the ground truth. A ROS-based LiDAR performance simulator [40] is employed to simulate point cloud noise in different fog levels by setting visibility. Since filtering discrete point cloud noise in adverser

weather is a well-solved problem in academia and industry, we only consider the quality degeneration of good points and filter out the noisy points. The RGB image quality degeneration is not involved in our experimental setting because image dehazing and enhancement are also a well-studied area. Figure 8 shows different objects under different fog level simulations at a distance of 15m in front of the LiDAR. On the IUSL dataset, the end-to-end 3D object detector is not integrated and we evaluate the pure online learning 3D object detector, because, to our knowledge, there is currently

Fig. 7 The mobile robot for data recording and LiDAR point cloud simulation under different fog levels

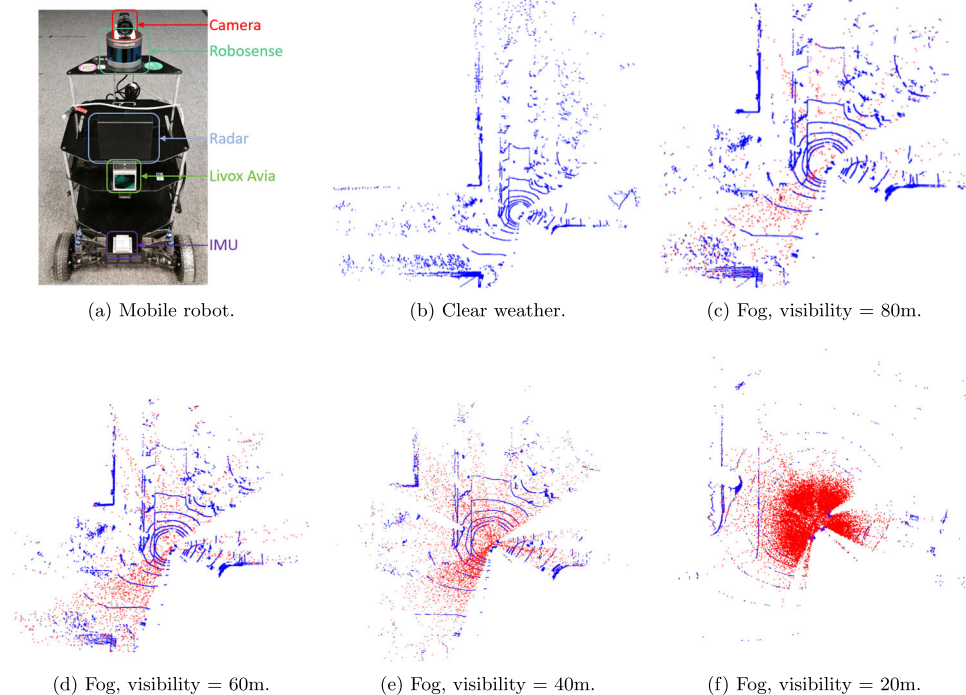
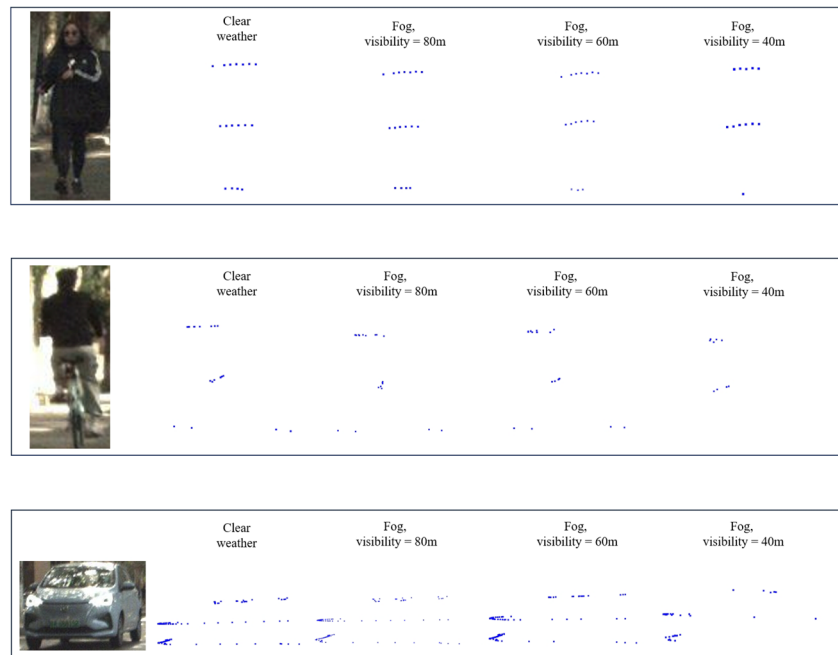


Fig. 8 Different objects under different fog level simulation at a distance of 15m in front of the LiDAR. When the visibility is 20m, all the object points disappear



no annotated dataset nor pre-trained model available for the 16-layer Robosense LiDAR.

For evaluation of the adaptation ability and knowledge retention ability, the testing steps are illustrated in Fig. 9. First, we evaluate the adaptation ability using *test-after-train* mode under clear weather conditions and with visibility of 80m, 60m, 40m, and 20m. Second, we directly test the best-performing model in the first step under different domains caused by visibility of 60m and 40m. We do not further test with visibility of 80m and 20m in fog, since, from Figs. 7 and 8, it can be seen that there is little LiDAR performance degeneration when the visibility is set to 80m, and the detector fails in too sparse point clouds when the visibility is set to 20m. Third, we load the best-performing model from the first step and execute the *test-then-train* evaluation [6, 24] using point clouds under visibility of 60m and 40m. Finally, following the *test-after-train* evaluation, we load the model from the third step and test its performance under clear weather conditions.

It's crucial to highlight that our model is trained without relying on any manual annotations. Instead, it learned directly from raw point clouds and the pre-trained YOLOv8 model 2D object detector.

CADC Dataset We use CADC dataset [10] to evaluate the domain adaptation and knowledge retention ability of our framework in the real world adverse weather conditions. The CADC dataset is the first autonomous vehicle dataset specifically designed to focus on adverse driving conditions. It was collected during the winter seasons of 2018 and 2019 in the Region of Waterloo, Canada, using data from eight cameras

and a 32-layer Velodyne LiDAR. The dataset divides driving sequences into four levels based on the number of snow points in the LiDAR point cloud to classify weather conditions: light snowfall (25-249), medium snowfall (250-499), heavy snowfall (500-749), and extreme snowfall (750-1500). Similar to the IUSL dataset, we only consider the quality degeneration of good points and filter out the snow points using DSOR [41]. Figure 10 shows the LiDAR points under light snowfall and extreme snowfall conditions. Given the absence of clear weather condition data in the CADC dataset, we selected all the sequences where the number of snow points does not exceed 100 from light snowfall conditions to train the PointPillars model. Additionally, we selected sequences where the average number of snow points exceed 900 from both heavy and extreme snowfall conditions to evaluate our method. Further details regarding the dataset information can be found in Table 2.

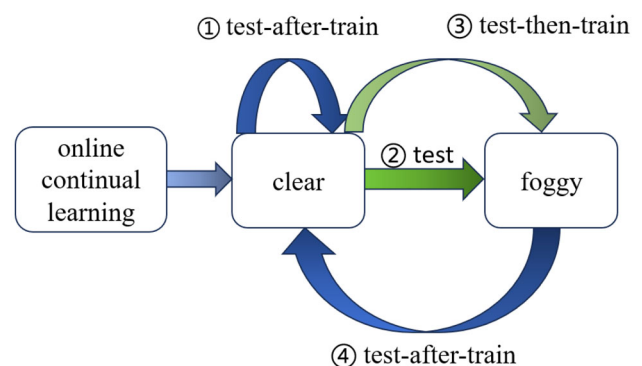
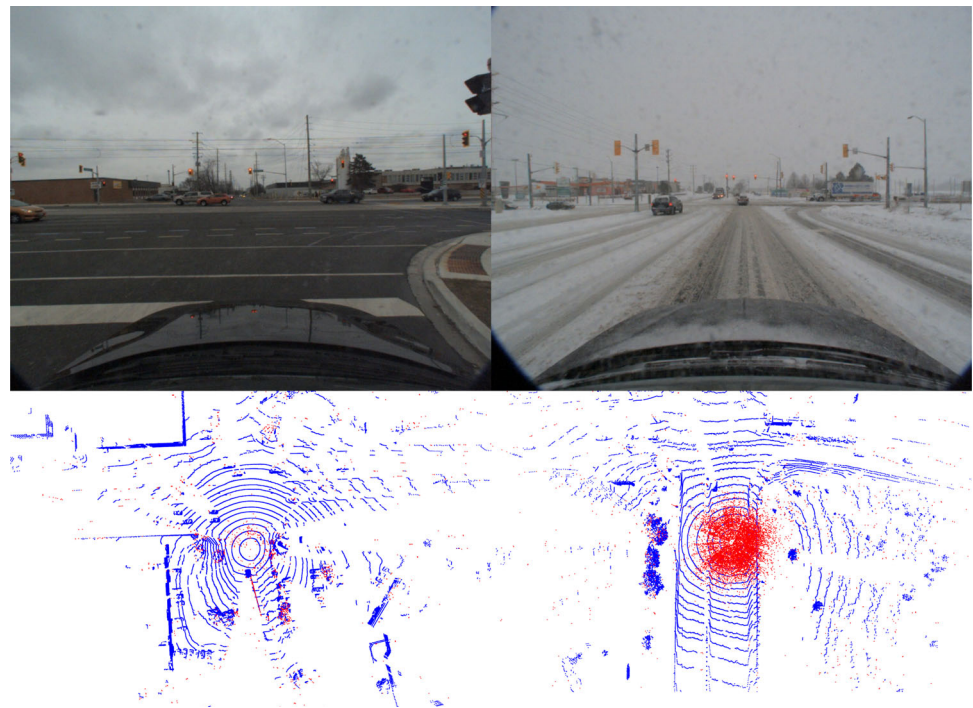


Fig. 9 Testing steps on the IUSL dataset

Fig. 10 LiDAR point clouds under light snowfall (left) and extreme snowfall (right) conditions, image for better visualization. Blue points represent good points, and red points represent snow points filtered by DSOR



4.2 Experiment Results

4.2.1 KITTI Dataset

On the KITTI dataset, we first evaluate the classification results of the 3D LiDAR-based online classifier. Since our approach doesn't depend on manually annotated labels, we compare the results of our online classifier to the YOLOv8 results used as a supervisor to help generate pseudo ground truth. The confusion matrix is shown in Fig. 11a.

It can be seen that our online classifier achieved considerably high accuracy, especially for car predictions when compared to the YOLOv8 results. However, the accuracy for

pedestrians and cyclists is relatively lower. The reason is that the YOLOv8 model pre-trained on the COCO dataset sometimes misses detecting bicycles. Consequently, our class mapping cannot merge classes of "person" and "bicycle" into "cyclist", this may lead our pseudo label generator to assign false classes to the object trajectories. Therefore, the classifier faces some difficulties in distinguishing between pedestrians and cyclists.

3D object detection is a joint result of both classification and localization. We let our online 3D object detector follow the *test-then-train* evaluation on the selected sequences. On the other hand, a fusion strategy, as detailed in Algorithm 1, is utilized to combine the results of the online 3D object detector and the PointPillars. We find that our proposed online 3D object detector improves the overall system performance compared to using the offline-trained end-to-end 3D object detector alone, whether using hand-crafted features or PointNet features, as measured by mean Average Precision(mAP). The comparative results are presented in Table 3. However, the AP for the car class is slightly lower compared to PointPillars. This disparity arises due to the sensitivity of Euclidean-based point cloud clustering to variations in point cloud density, which results in an object's cluster being segmented into multiple ones, especially when the point cloud is sparse.

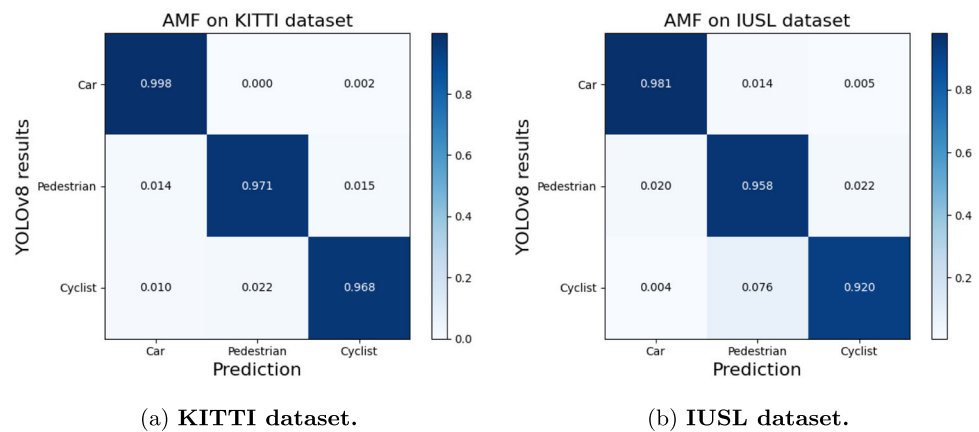
As shown in Fig. 12, the point cloud of a van has been segmented into two clusters and incorrectly classified as a car due to misleading pseudo labels provided by YOLOv8. This over-segmentation leads to false positive classifications in our final detection results. However, our method has a better

Table 2 Selected sequences from CADC dataset

Date	Drive	Weather conditions	Number
2018_03_06	0001	Light snowfall	26
2018_03_06	0010	Light snowfall	75
2018_03_07	0004	Light snowfall	42
2018_03_07	0006	Light snowfall	42
2018_03_07	0005	Light snowfall	100
2019_02_27	0013	Extreme snowfall ¹	527
2019_02_27	0051	Extreme snowfall	1105
2019_02_27	0068	Extreme snowfall	1327

¹Considering the average number of snow points for drives 0013, 0051, and 0068 from 2019_02_27 exceed 900, we also classify drive 0013 as extreme snowfall, despite it originally being categorized as heavy snowfall

Fig. 11 The confusion matrix of our online classifier on the KITTI and IUSL dataset



3D object detection performance for smaller objects, e.g., pedestrians and cyclists, since these objects don't suffer from the above issue. Moreover, clustering-based object detection can detect any obstacles. Compared with the batch-based online learning method fusing with PointPillars, which uses deep learning features and a few-shot learning strategy, our approach shows better results for both hand-crafted features and deep learning features and needn't maintain a memory bank of samples, thanks to the efficient learning capability of AMF.

4.2.2 IUSL Dataset

The confusion matrix of classification results of the 3D online classifier under clear weather conditions is shown in Fig. 11b. For the IUSL dataset, the online classifier exhibits relatively lower classification accuracy compared to the KITTI dataset, especially for the cyclist class, because sparse point clouds of the 16-layer LiDAR make it challenging to distinguish between the cyclist and pedestrian clusters.

To evaluate the domain adaptation and knowledge retention ability, we follow the evaluation steps as shown in Fig. 9. Figure 13 depicts a typical case showing the variation in cross-domain detection performance of our online 3D object detector when the target domain visibility is set to 40m. It can be seen that deploying a model trained in the clear weather,

i.e., source domain, directly to the foggy weather, i.e., target domain, leads to a decline in detection performance. However, following the application of online continual learning, the detection performance improved rapidly. Furthermore, the model maintains high detection performance in the source domain of clear weather, experiencing only a slight degradation.

Peak Performance We deploy the online 3D object detector under various weather conditions and iterate the model training for five epochs, following the first step in the evaluation pipeline. After completing training, we save the best-performing models under various weather conditions. Subsequently, we evaluate these models' performance under the weather conditions corresponding to their training settings. The experimental results are presented in Table 4, which can be regarded as the peak performance of the online 3D object detector in the absence of using ground truth.

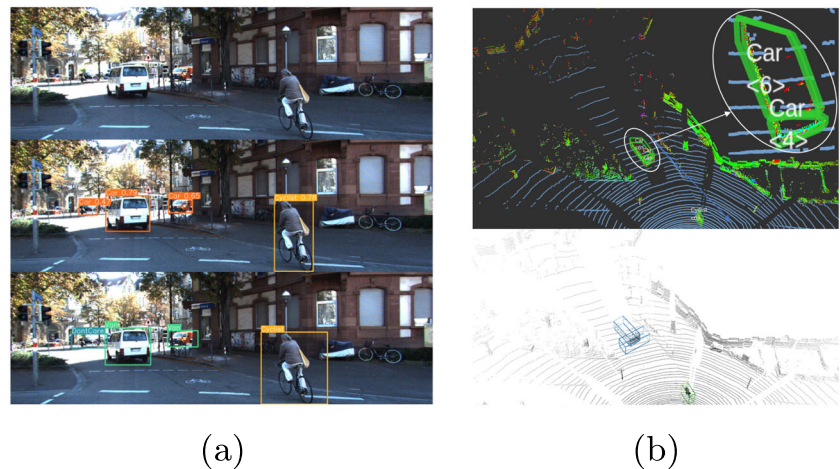
Adaptation Performance In order to demonstrate the domain adaptation ability of our online 3D object detector, we follow the second and third steps in the evaluation pipeline. First, we directly deploy the peak performance model trained under clear weather conditions to foggy weather conditions. Then, we adopt *test-then-train* evaluation mode, where the online 3D object detector first performs detection and then undergoes online training. The results are shown in Table 5. It can

Table 3 The results of our framework on KITTI dataset. The bold entries highlight the best detection performance for each class

Method	Car			Pedestrian			Cyclist			mAP
	Easy	Mod.	Hard	Easy	Mod.	Hard	Easy	Mod.	Hard	
PointPillars [13]	96.51	86.97	86.89	59.55	52.04	51.94	66.85	59.10	58.32	65.70
Few-shotOL [9]	94.38	85.81	85.79	58.42	57.89	51.02	73.21	64.85	58.01	69.52
Ours(H) ¹	92.11	84.59	84.64	57.77	58.64	58.65	74.58	66.21	58.68	69.81
Ours(P) ²	94.45	85.84	85.85	59.51	59.79	59.72	74.23	66.00	58.68	70.54

¹H represents the usage of Hand-crafted features. ²P represents the usage of PointNet features

Fig. 12 On the left side from top to bottom: the raw image, YOLOv8 results, and the ground truth. YOLOv8 struggles to distinguish between a car and a van. On the right side from top to bottom: the point cloud clustering results, and the final detections given by our approach



be seen that when directly deploying the peak performance model trained in the first step to the target domains, the online 3D object detector experiences a degradation in detection performance. Specifically, when utilizing hand-crafted features, the degradation is more pronounced compared to using PointNet features. However, upon deploying our online learning framework, there is a significant improvement in the detection performance, which proves that our approach possesses strong domain adaptation capabilities.

Retention Performance To demonstrate the retention ability of our method for past learned knowledge, we follow the fourth step in the evaluation pipeline. First, we load the model from the third step and test its performance in clear weather using *test-after-train* evaluation mode. The experimental results, detailed in Table 6, unveil a noteworthy trend. It is evident that detection performance tested in the clear weather experiences a degradation after learning successively from LiDAR point clouds under both clear weather conditions and foggy weather conditions due to catastrophic forgetting in machine learning. However, this degradation is

remarkably minimal at this stage. Additionally, the model trained with hand-crafted features shows a slightly smaller performance degradation compared to the one trained with PointNet features. This suggests that our method effectively retains a substantial portion of its performance from the first domain when sequentially learning from two different domains' data, showcasing its ability to effectively overcome catastrophic forgetting. Remarkably, there is an unexpected twist. Upon learning from LiDAR point clouds under the visibility of 60m in fog, using hand-crafted features, the model's performance actually demonstrates improvement in terms of mAP. This is possible because hand-crafted features have correctly extracted the intrinsic similar features from LiDAR point clouds under the visibility of 60m, resembling those present under clear weather conditions.

4.2.3 CADC Dataset

On CADC dataset, we use the PointPillars as the offline-trained end-to-end 3D object detector and follow the test steps as shown in Fig. 9. The evaluation results of our framework can be found in Table 7. In experiment [a], we test the point-pillars on the drive 0005 (not used for training) from date 2018_03_07 after training it on the rest of the drives under light snowfall weather conditions. Additionally, in experiment [b], we follow the *test-then-train* evaluation mode to test our framework. It can be seen that our method significantly outperforms the offline-trained 3D object detector (PointPillars), indicating that our online 3D object detector can effectively improve the system's performance. Experiment [c] and experiment [d] showcase the domain adaptation ability of our framework. Specially: 1) In experiment [c], we directly test the offline-trained PointPillars in extreme snowfall weather conditions; 2) In experiment [d], we first load

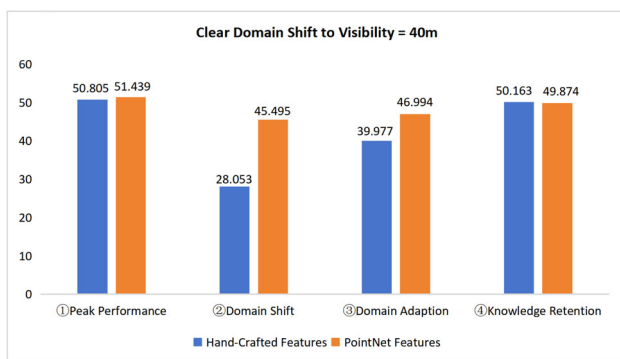


Fig. 13 The typical results of cross-domain experiments when the visibility is set to 40m

Table 4 Peak performance of the online 3D object detector

Features ¹	Source domain ²	Target domain ²	Car	Pedestrian	Cyclist	mAP
Hand-crafted	Clear weather	Clear weather	42.902	40.821	68.691	50.805
Hand-crafted	Visibility = 80m	Visibility = 80m	42.199	41.468	70.188	51.285
Hand-crafted	Visibility = 60m	Visibility = 60m	41.150	41.062	69.593	50.602
Hand-crafted	Visibility = 40m	Visibility = 40m	31.294	44.794	68.343	48.144
Hand-crafted	Visibility = 20m	Visibility = 20m	0.000	29.708	42.162	23.957
PointNet	Clear weather	Clear weather	42.210	43.485	68.621	51.439
PointNet	Visibility = 80m	Visibility = 80m	44.556	44.766	69.327	52.883
PointNet	Visibility = 60m	Visibility = 60m	42.494	44.088	69.776	52.119
PointNet	Visibility = 40m	Visibility = 40m	29.509	45.012	69.487	48.003
PointNet	Visibility = 20m	Visibility = 20m	0.000	29.049	35.627	21.559

¹Hand-crafted represents using hand-crafted features, PointNet represents using PointNet features ²Source domain refers to the specific weather conditions in which the online 3D object detector has been trained on LiDAR point cloud data. Target domain refers to the weather conditions under which the online 3D object detector is deployed

The bold entries highlight the best detection performance for each class

Table 5 Results of domain adaptation on IUSL dataset

Features ¹	Source domain ²	Target domain ²	Car	Pedestrian	Cyclist	mAP
Hand-crafted	Clear weather	Visibility = 60m	42.395	24.579	58.397	41.790
Hand-crafted†	Clear weather	Visibility = 60m	44.327	45.486	57.997	49.270
Improvement	Clear weather	Visibility = 60m	1.932	20.907	−0.400	7.480
Hand-crafted	Clear weather	Visibility = 40m	23.645	17.455	43.061	28.053
Hand-crafted†	Clear weather	Visibility = 40m	27.288	44.888	47.756	39.977
Improvement	Clear weather	Visibility = 40m	3.643	27.433	4.695	11.924
PointNet	Clear weather	Visibility = 60m	43.676	43.688	68.450	51.938
PointNet†	Clear weather	Visibility = 60m	43.179	47.385	69.448	53.337
Improvement	Clear weather	Visibility = 60m	−0.497	3.697	0.998	1.399
PointNet	Clear weather	Visibility = 40m	25.527	42.951	68.006	45.495
PointNet†	Clear weather	Visibility = 40m	26.949	45.959	68.073	46.994
Improvement	Clear weather	Visibility = 40m	1.422	3.008	0.067	1.499

¹Hand-crafted represents the Hand-crafted features module, PointNet represents the PointNet features module, while † represents the usage of *test-then-train* evaluation mode. ²Source domain refers to the specific weather conditions in which the online 3D object detector has been trained on LiDAR point cloud data. Target domain refers to the weather conditions under which the online 3D object detector is deployed

Table 6 Results of retention performance on the IUSL dataset

Features ¹	Source domain ²	Target domain ²	Car	Pedestrian	Cyclist	mAP
Hand-crafted	Clear weather	Clear weather	42.902	40.821	68.691	50.805
Hand-crafted	+ visibility = 60m	Clear weather	41.641	42.968	68.316	50.975
Degradation	+ visibility = 60m	Clear weather	1.261	−2.147	0.375	−0.17
Hand-crafted	+ visibility = 40m	Clear weather	37.470	43.322	69.697	50.163
Degradation	+ visibility = 40m	Clear weather	5.432	−2.501	−1.006	0.642
PointNet	Clear weather	Clear weather	42.210	43.485	68.621	51.439
PointNet	+ visibility = 60m	Clear weather	41.843	42.194	69.386	51.141
Degradation	+ visibility = 60m	Clear weather	0.367	1.291	−0.765	0.298
PointNet	+ visibility = 40m	Clear weather	40.933	40.331	68.358	49.874
Degradation	+ visibility = 40m	Clear weather	1.277	3.154	0.263	1.565

¹Hand-crafted represents hand-crafted features module, PointNet represents PointNet features module ²Source domain refers to the specific weather conditions in which the online 3D object detector has been trained on LiDAR point cloud data. Target domain refers to the weather conditions under which the online 3D object detector is deployed

Table 7 CADC dataset results

	Method ¹	Source domain	Target domain	Car	Pedestrian	mAP ²
[a]	PointPillars	Light snowfall	Light snowfall	87.743	39.314	63.529
[b]	Ours	Light snowfall	Light snowfall	90.455	88.923	89.689
[c]	PointPillars	Light snowfall	Extreme snowfall	48.053	35.208	41.631
[d]	Ours	Light snowfall	Extreme snowfall	47.475	44.425	45.950
[e]	Ours	Light snowfall + extreme snowfall	Light snowfall	90.244	89.736	89.990

¹Our method uses PointNet features. ²Since there are very few annotations for Bicycles and Trucks in CADC dataset, we focus our evaluation on the categories of Cars and Pedestrians

Table 8 Efficiency of different modules

	Module	Running time	CPU usage	GPU usage
[a]	YOLOv8	17ms	36.2%	2.3GB
[b]	Ground segmentation	7ms	19.5%	—
[c]	Point cloud clustering*	22ms	52.4%	—
[d]	2D-3D association*	0.28ms	—	—
[e]	Feature extractor(P)	13ms	22.7%	0.8GB
[f]	Feature extractor(H)	2ms	1.3%	—
[g]	Muti-object tracker†	0.29ms	1.3%	—
[h]	Sample annotator†	0.04ms	—	—
[i]	Online classifier(testing thread)	5ms	12.7%	—
[j]	Online classifier(training thread)	18ms	20.9%	—
[k]	PointPillars	24ms	33.1%	2.1GB
sum ¹	Ours(P)	88.61ms	177.9%	5.2GB
sum ¹	Ours(H)	77.61ms	156.5%	4.4GB

¹The efficiency of the online classifier's training thread does not need to be taken into account because testing and training occur in two independent threads. A slower training thread will not block the testing thread. Module with the same symbol means implemented in the same ROS node, the CPU usage is provided in total

Table 9 Scalability of point cloud clustering module with different resolution LiDARs

LiDAR sensor	Voxel size of down sample	Running time	CPU usage
Robosense 16 layer	0.10	5ms	11.4%
Velodyne 32 layer	0.10	24ms	52.4%
Velodyne 64 layer	0.25	22ms	38.7%

the model of our online 3D object detector that was online-trained in experiment [b], and follow the *test-then-train* evaluation mode. Experiment [e] demonstrates retention ability of our method for past learned knowledge. We load the model of the online 3D object detector from experiment [d] and test its performance in light snowfall using the *test-after-train* evaluation mode. It can be seen that our method still maintains high detection performance and does not show-case catastrophic forgetting.

4.2.4 Efficiency and Scalability

Efficiency We provide the running time, CPU usage and GPU usage of different modules for our framework on CADC dataset. All the experiments are conducted with an Intel i7-12700K CPU and a NVIDIA GeForce RTX 3090 GPU. The details can be found in Table 8. It is important to note that our online classifier contains two threads, one for testing and the other for training. The training thread updates the model at a slower rate (about 2Hz in our experiments), depending on whether the label generator component has produced reliable learning samples. The testing thread processes streaming data in a callback manner.

Scalability Our method has been tested with three types of LiDAR sensors: Velodyne 64-layer LiDAR, Velodyne 32-layer LiDAR, and Robosense 16-layer LiDAR. These sensors provide varying numbers of LiDAR points, which directly affect the efficiency of our framework, especially the Point Cloud Clustering module. The running time and CPU usage of the Point Cloud Clustering module with different LiDAR sensors are detailed in Table 9. Although our method meets real-time requirements for 64-layer, 32-layer, and 16-layer LiDARs, we must acknowledge that Euclidean-based point cloud clustering method may not satisfy the real-time demands of higher resolution LiDARs. However, range image-based clustering method, such as depth clustering [42], are not sensitive to the number of LiDAR layers. Thanks to our ROS-based framework, we have the flexibility to integrate these methods in future work, ensuring scalability with higher resolution LiDARs.

5 Conclusions

In this work, we proposed an approach that aims to address the unsolved yet challenging task of semi-supervised online continual learning for 3D object detection in autonomous driving and mobile robotics. By integrating the 3D multi-object tracker and the sample annotator modules, the framework automatically generated pseudo labels for each object trajectory by leveraging a pre-trained 2D object detector.

With the innovative integration of the AMF, our method can efficiently learn from the unlabelled LiDAR point clouds and make detections in conjunction with the point cloud clustering. Additionally, due to the efficiency and accuracy of pseudo labels generation, coupled with the rapid adaptation capability of AMF to changes in data distribution, our approach quickly adapted to LiDAR performance degeneration while retaining the knowledge acquired in clear weather.

Future work involves integrating 4D radar into the online continual learning framework.

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Data Availability The IUSL dataset is available at <https://github.com/npu-ius-lab/OCL3D>.

Code Availability The source code is available at <https://github.com/npu-ius-lab/OCL3D>.

Declarations

Conflicts of Interest The authors declare no conflict of interest.

Ethics Approval This article does not contain any studies with human participants or animals performed by any of the authors.

Consent for participate All authors of this research paper have consented to participate in the research study.

Consent for Publication All authors and involved researchers give their consent for the publication of the present paper to be published in the Journal of Intelligent and Robotic Systems.

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Binhong Liu received a Bachelor of Engineering degree from Beijing Forestry University in 2022. He is currently pursuing a Ph.D. degree in Unmanned Systems Science and Technology at the Unmanned System Research Institute, Northwestern Polytechnical University, Xi'an, China. His research interests include robotics, 3D object detection, continual learning, place recognition, and point cloud registration.

Dexin Yao received a Bachelor of Engineering degree from Northwest A&F University in 2021. From 2021 to 2024, he pursued an M.S. degree in Electronics and Information at the Unmanned System Research Institute, Northwestern Polytechnical University, Xi'an, China. His research interests include robotics, 3D object detection, and few-shot learning.

Rui Yang received the M.S. degree in computer science and technology from Beijing Technology and Business University (BTBU), Beijing, China, in 2019. He is pursuing the Ph.D. degree with the Distributed Artificial Intelligence and Knowledge Laboratory (CIAD), University of Technology of Belfort-Montbéliard (UTBM), Belfort, France., His current research interests include robot perception and learning for autonomous driving.

Zhi Yan received the Ph.D. degree from the University of Paris 8, France, in 2012., From 2013 to 2015, he was a Postdoctoral Research Fellow with the CAR Team, IMT Lille Douai, France. From 2016 to 2017, he was also a Postdoctoral Research Fellow with the Lincoln Centre for Autonomous Systems (L-CAS), University of Lincoln, U.K., working on the Horizon 2020 Project FLOBOT. Since 2017, he has been an Assistant Professor (Maître de conférences) with the Distributed Artificial Intelligence and Knowledge Laboratory (CIAD), University of Technology of Belfort-Montbéliard (UTBM). His research interests are in autonomous driving, mobile robotics, and chronorobotics.

Tao Yang received the B.S. degree in detection, guidance, and control technology and the M.S. degree in control engineering from Northwestern Polytechnical University (NPU), Xi'an, China, in 2011 and 2014, respectively, and the Ph.D. degree in automation from the University of Technology of Belfort-Montbéliard (UTBM), Belfort, France, in 2019. From 2019 to 2020, he was a Postdoctoral Research Fellow with the Distributed Artificial Intelligence and Knowledge Laboratory (CIAD), UTBM. Since 2020, he has been an Associate Professor with the Unmanned System Research Institute (USRI), NPU. His research interests include computer vision, point cloud processing, and chronorobotics.